

III. POLYHARMONIC SPLINE INTERPOLATION

It is remarkable that, despite the fact that it is continuous up to the second derivatives, the trispline interpolation fails so badly at reproducing the Laplacian of the electron density, which ultimately precludes its use in a Newton-based CP search algorithm.

The method proposed in this work is based on polyharmonic spline interpolation⁶¹⁻⁶⁴ combined with a smoothing transformation. Polyharmonic splines combine optimal smoothing properties with flexible stencil selection.⁶⁵ A polyharmonic spline interpolant is built as the sum of a low-degree polynomial plus,

$$\rho(\mathbf{x}) = \sum_i^N w_i \phi(|\mathbf{x} - \mathbf{x}_i|) + p(\mathbf{x}), \quad (12)$$

where $\{\mathbf{x}_i\}$ are the positions of the N grid nodes used in the interpolation stencil. The polynomial term is written as

$$p(\mathbf{x}) = \mathbf{v} \cdot (1, \mathbf{x})^T. \quad (13)$$

The $\mathbf{w} = \{w_i\}$ and $\mathbf{v} = (v_0, v_x, v_y, v_z)$ are the $N + 4$ interpolation coefficients, which are determined by applying the following $N + 4$ constraints:

$$\rho(\mathbf{x}_i) = \rho_i, \quad (14)$$

$$\sum_i w_i = 0, \quad (15)$$

$$\sum_i w_i \mathbf{x}_i = 0, \quad (16)$$

where the sums run over the stencil grid nodes. The application of these constraints leads directly to the following matrix equation:

$$\Phi(\mathbf{w} \quad \mathbf{v})^T = \begin{pmatrix} F & C \\ C^T & 0 \end{pmatrix} (\mathbf{w} \quad \mathbf{v})^T = \begin{pmatrix} \boldsymbol{\rho}^T \\ 0 \end{pmatrix}, \quad (17)$$

where $\boldsymbol{\rho} = (\rho_1, \dots, \rho_N)$ is the vector of electron densities at the stencil grid nodes. The symmetric F matrix is defined as

$$F_{ij} = \phi(|\mathbf{x}_i - \mathbf{x}_j|), \quad (18)$$

and row i in matrix C is

$$C_i = (1, \mathbf{x}_i). \quad (19)$$

The radial function for a polyharmonic spline of degree 2 in three dimensions, $\phi(r) = r^3$, was used. Polyharmonic splines are commonly used for the interpolation of scattered data.

Figure 1 shows clearly that the main difficulty for the interpolants lies in accurately reconstructing the electron density in the regions close to the nuclei, where $\rho(\mathbf{x})$ changes very quickly. In this work, instead of using the electron density directly, the interpolation is carried out on the function resulting from the smoothing transformation,

$$f(\mathbf{x}) = \ln \left(\frac{\rho(\mathbf{x})}{\rho_0(\mathbf{x})} \right), \quad (20)$$

where $\rho_0(\mathbf{x})$ is the promolecular density, the sum of atomic *in vacuo* densities at the system geometry. (Note there is an arbitrariness⁶⁶ in the choice of reference atomic densities for building ρ_0 , but we expect any reasonable reference density will do.) In the critic2 implementation, $\rho_0(\mathbf{x})$ is known only numerically, but because it is calculated as a sum of contributions from one-dimensional radial functions, its value and its derivatives at an arbitrary point can be calculated very accurately using one-dimensional interpolation. The deformation density $[\rho(\mathbf{x}) - \rho_0(\mathbf{x})]$, which measures deviations from the promolecular density, is a popular chemical bonding indicator because density accumulates in the interatomic region upon the formation of a chemical bond. The function in Eq. (20) also expresses deviations with respect to the promolecular density. Since the changes induced in the electron density by the formation of chemical bonds are relatively minor compared to the density itself and happen mostly in the interatomic regions, where the density varies slowly, $f(\mathbf{x})$ is an excellent candidate for a smoothing transformation.

The interpolant built from the polyharmonic spline [Eq. (13)] applied to the smooth density function [Eq. (20)] has analytical first and second derivatives that can be readily calculated,

$$\tilde{\rho} = \rho_0 \times e^f, \quad (21)$$

$$\tilde{\rho}_\xi = \rho \times \left(f_\xi + \frac{\rho_{0\xi}}{\rho_0} \right), \quad (22)$$

$$\tilde{\rho}_{\xi\zeta} = \rho \times \left(f_{\xi\zeta} + \frac{\rho_\xi \rho_\zeta}{\rho^2} + \frac{\rho_{0\xi} \rho_{0\zeta}}{\rho_0} - \frac{\rho_{0\xi} \rho_{0\zeta}}{\rho_0^2} \right), \quad (23)$$

where the ξ and ζ subscripts denote the derivatives with respect to the Cartesian coordinates (x, y, z) of the corresponding functions. The derivatives of the smooth function are obtained as the derivatives of the polyharmonic spline,

$$f = \sum_i w_i \phi(r) + \mathbf{v} \cdot \mathbf{x} + v_0, \quad (24)$$

$$f_\xi = \sum_i w_i \phi'(r) \frac{\mathbf{x}_\xi - \mathbf{x}_{i\xi}}{r} + v_\xi, \quad (25)$$

$$f_{\xi\zeta} = \sum_i w_i \left[\phi''(r) \frac{(\mathbf{x}_\xi - \mathbf{x}_{i\xi})(\mathbf{x}_\zeta - \mathbf{x}_{i\zeta})}{r^2} + \phi'(r) \left(\frac{\delta_{\xi\zeta} r}{r} - \frac{(\mathbf{x}_\xi - \mathbf{x}_{i\xi})(\mathbf{x}_\zeta - \mathbf{x}_{i\zeta})}{r^3} \right) \right], \quad (26)$$

where $\mathbf{x}_\xi$ is the ξ component of vector $\mathbf{x}$, $r = |\mathbf{x} - \mathbf{x}_i|$, and $\phi'(r) = 3r^2$ and $\phi''(r) = 6r$ are the first and second derivatives of the radial function.

In this work, the number of radial basis functions used is $N = 8^3 = 512$. For a given target point $\mathbf{x}$, the N grid nodes on which these functions are based are the closest in Euclidean distance to the nearest grid node $\mathbf{x}_0$ (the "base" node). The use of a base node, instead of simply taking the N grid nodes closest to the target point, is important because the stencil geometry and the Φ matrix in Eq. (17) are the same for every target point, which means they can be pre-computed and stored. More importantly, the fact that the stencil nodes are determined by Euclidean distance, rather than by linear combinations of the grid basis vectors, eliminates the lattice bias from the interpolation method since the same grid with two different bases would give the same stencil for a particular target point.

As the target point moves from one base node to another, there is a change in the stencil, and therefore the interpolant obtained from simply using $\tilde{\rho}$ [Eq. (21)] is not continuous. However, this discontinuity is minor because adjacent base nodes share a large part of their stencils. Therefore, in order to address this problem, we build our interpolation function as a linear combination of the interpolants with base nodes in the vicinity of the target point. Specifically, if $\mathbf{x}$ is the target point, we assign a weight to each grid node $\mathbf{x}_i$ given by

$$w_i = \begin{cases} \exp\left(\frac{d_i^3}{a^3(d_i^3 - a^3)}\right) & \text{if } 0 \leq d_i < a, \\ 0 & \text{if } d_i \geq a, \end{cases} \quad (27)$$

where $d_i = |\mathbf{x} - \mathbf{x}_i|$ and a is a range parameter, which in this work is taken as twice the longest grid step. This weight function has a maximum at $d = 0$ and goes smoothly to zero at $x = a$ with continuous first and second derivatives. The density is then interpolated as

$$\rho(\mathbf{x}) = \frac{\sum_i w_i \tilde{\rho}_i(\mathbf{x})}{\sum_i w_i}, \quad (28)$$

where $\tilde{\rho}_i(\mathbf{x})$ is the density predicted by the interpolant based on node grid i . The first and second derivatives required for the CP location are calculated straightforwardly from Eqs. (27) and (28). This choice of weight function makes the proposed interpolation method have continuous derivatives up to at least second order and be free of lattice bias.

The dramatic improvement in the accuracy of the interpolation using the new method can be seen in the phenol dimer example [Fig. 2 (right)]. The spurious oscillations in the density and its derivatives have disappeared, and the interpolated values are in exceptional agreement with the analytical density, except for minor disagreements very close to the nuclei and in the very steep shell structure of the Laplacian. In Sec. IV, it is shown that the CP topology of a molecule or solid can be recovered when this interpolant is used in combination with Newton's method, yielding accurate properties at the critical points and molecular graphs that are visually indistinguishable from their analytical counterparts.

Regarding the practical details, as mentioned earlier, the fact that the stencil geometry is independent of the target point position

allows an efficient implementation. The Φ matrix, the stencil integer offsets, and the promolecular densities and their derivatives on the grid are precomputed and stored. [In fact, Φ could be saved in the decomposed form to speed up the solution of the system of linear equations in Eq. (17).] A single evaluation of the interpolant at an arbitrary target point requires, for every interpolant $\tilde{\rho}_i$ appearing in Eq. (28), the retrieval of the N density values on the grid nodes belonging to the corresponding stencil [right-hand side of Eq. (17)], then solving the system to obtain the interpolation coefficients, and finally applying Eqs. (21)–(26) to calculate the interpolated density and its derivatives. The cost of the interpolant evaluation is independent of the grid size and the system size, although the previously mentioned preparatory calculations are not. In addition, CP location based Newton's method is trivially parallelizable, as Newton searches launched from different seeds are independent of each other.